KAIZEN MATHS

Lesson starter

Midpoint and Length of Line Segments

Challenge

Name _____

S1

A (4, -1) and B (-1, 5) are endpoints of a line segment. Find the midpoint and the length of AB.

S2

AB is the diameter of a circle. A (2, -5) and B (0, -1) . Find the centre and radius.

S3

A (-4, 0) and B (3, 2) are endpoints of a line segment. Find the midpoint and the length of AB.

S4

AB is the diameter of a circle. A (-1, 1) and B (3, -2) . Find the centre and radius.

Show your working

KAIZEN MATHS

Lesson starter

Midpoint and Length of Line Segments

Challenge

Name _____

S1

A (4, -1) and B (-1, 5) are endpoints of a line segment. Find the midpoint and the length of AB.

S2

AB is the diameter of a circle. A (2, -5) and B (0, -1) . Find the centre and radius.

S3

A (-4, 0) and B (3, 2) are endpoints of a line segment. Find the midpoint and the length of AB.

S4

AB is the diameter of a circle. A (-1, 1) and B (3, -2) . Find the centre and radius.

GEO-3384-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Challenge

Name _____

E1

A (3, 3) and B (-5, -2) are endpoints of a line segment. Find the midpoint and the length of AB.

E2

AB is the diameter of a circle. A (-3, -3) and B (-1, 4) . Find the centre and radius.

E3

A (1, 3) and B (2, 5) are endpoints of a line segment. Find the midpoint and the length of AB.

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Challenge

Name _____

E1

A (3, 3) and B (-5, -2) are endpoints of a line segment. Find the midpoint and the length of AB.

E2

AB is the diameter of a circle. A (-3, -3) and B (-1, 4) . Find the centre and radius.

E3

A (1, 3) and B (2, 5) are endpoints of a line segment. Find the midpoint and the length of AB.

GEO-3384-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Challenge

Name _____

E1

A (3, 3) and B (-5, -2) are endpoints of a line segment. Find the midpoint and the length of AB.

E2

AB is the diameter of a circle. A (-3, -3) and B (-1, 4) . Find the centre and radius.

E3

A (1, 3) and B (2, 5) are endpoints of a line segment. Find the midpoint and the length of AB.

GEO-3384-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Challenge

Name _____

E1

A (3, 3) and B (-5, -2) are endpoints of a line segment. Find the midpoint and the length of AB.

E2

AB is the diameter of a circle. A (-3, -3) and B (-1, 4) . Find the centre and radius.

E3

A (1, 3) and B (2, 5) are endpoints of a line segment. Find the midpoint and the length of AB.

GEO-3384-L4 · E

Show your working

Teacher answers & working

Midpoint and Length of Line Segments · Challenge

GEO-3384-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

A (4, -1) and B (-1, 5) are endpoints of a line segment. Find the midpoint and the length of AB.

Answer:

Midpoint $(\frac{3}{2}, 2)$; length $\sqrt{61}$ units

Working:

First find the midpoint by averaging coordinates.

$M = (\frac{4+-1}{2}, \frac{-1+5}{2}) = (\frac{3}{2}, 2)$

Then find the length using the distance formula.

$AB = \sqrt{((-1 - (4))^2 + (5 - (-1))^2)}$

$AB = \sqrt{(-5^2 + 6^2)} = \sqrt{61}$

So the midpoint is $(\frac{3}{2}, 2)$ and AB = $\sqrt{61}$ units.

S2

AB is the diameter of a circle. A (2, -5) and B (0, -1) . Find the centre and radius.

Answer:

Centre (1, -3); radius $\sqrt{5}$

Working:

The centre of the circle is the midpoint of the diameter.

Centre = $(\frac{2+0}{2}, \frac{-5+-1}{2})$ = (1, -3)

The radius is half the diameter. First find the length of AB.

$AB = \sqrt{((0 - (2))^2 + (-1 - (-5))^2)} = \sqrt{20}$

Radius = $\frac{AB}{2} = \frac{\sqrt{20}}{2}$

Therefore the centre is (1, -3) and the radius is $\sqrt{5}$.

S3

A (-4, 0) and B (3, 2) are endpoints of a line segment. Find the midpoint and the length of AB.

Answer:

Midpoint $(\frac{-1}{2}, 1)$; length $\sqrt{53}$ units

Working:

First find the midpoint by averaging coordinates.

$M = (\frac{-4+3}{2}, \frac{0+2}{2}) = (\frac{-1}{2}, 1)$

Then find the length using the distance formula.

$AB = \sqrt{((3 - (-4))^2 + (2 - (0))^2)}$

$AB = \sqrt{(7^2 + 2^2)} = \sqrt{53}$

So the midpoint is $(\frac{-1}{2}, 1)$ and AB = $\sqrt{53}$ units.

Teacher answers & working

Midpoint and Length of Line Segments · Challenge

GEO-3384-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

AB is the diameter of a circle. A (-1, 1) and B (3, -2) . Find the centre and radius.

Answer:

Centre (1, $\frac{-1}{2}$); radius $\frac{\sqrt{25}}{2}$

Working:

The centre of the circle is the midpoint of the diameter.

Centre = ($\frac{-1+3}{2}$, $\frac{1+-2}{2}$) = (1, $\frac{-1}{2}$)

The radius is half the diameter. First find the length of AB.

$AB = \sqrt{((3 - (-1))^2 + (-2 - (1))^2)} = \sqrt{25}$

Radius = $\frac{AB}{2}$ = $\frac{\sqrt{25}}{2}$

Therefore the centre is (1, $\frac{-1}{2}$) and the radius is $\frac{\sqrt{25}}{2}$.

E1

A (3, 3) and B (-5, -2) are endpoints of a line segment. Find the midpoint and the length of AB.

Answer:

Midpoint (-1, $\frac{1}{2}$); length √89 units

Working:

First find the midpoint by averaging coordinates.

$M = (\frac{3+-5}{2}, \frac{3+-2}{2}) = (-1, \frac{1}{2})$

Then find the length using the distance formula.

$AB = \sqrt{((-5 - (3))^2 + (-2 - (3))^2)}$

$AB = \sqrt{(-8^2 + -5^2)} = \sqrt{89}$

So the midpoint is (-1, $\frac{1}{2}$) and AB = √89 units.

E2

AB is the diameter of a circle. A (-3, -3) and B (-1, 4) . Find the centre and radius.

Answer:

Centre (-2, $\frac{1}{2}$); radius $\frac{\sqrt{53}}{2}$

Working:

The centre of the circle is the midpoint of the diameter.

Centre = ($\frac{-3+-1}{2}$, $\frac{-3+4}{2}$) = (-2, $\frac{1}{2}$)

The radius is half the diameter. First find the length of AB.

$AB = \sqrt{((-1 - (-3))^2 + (4 - (-3))^2)} = \sqrt{53}$

Radius = $\frac{AB}{2}$ = $\frac{\sqrt{53}}{2}$

Therefore the centre is (-2, $\frac{1}{2}$) and the radius is $\frac{\sqrt{53}}{2}$.

Teacher answers & working

Midpoint and Length of Line Segments · Challenge

GEO-3384-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

A (1, 3) and B (2, 5) are endpoints of a line segment. Find the midpoint and the length of AB.

Answer:

Midpoint $(\frac{3}{2}, 4)$; length √5 units

Working:

First find the midpoint by averaging coordinates.

$M = (\frac{1+2}{2}, \frac{3+5}{2}) = (\frac{3}{2}, 4)$

Then find the length using the distance formula.

$AB = \sqrt{((2 - (1))^2 + (5 - (3))^2)}$

$AB = \sqrt{(1^2 + 2^2)} = \sqrt{5}$

So the midpoint is $(\frac{3}{2}, 4)$ and AB = √5 units.